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ANJANA SANTOSH KUMAR , M.S.

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Personal Summary

Graduate researcher with a strong foundation in transcriptomic analysis, neurodegenerative disease models, and data-driven neuroscience. Experienced with single-nucleus RNA-seq, gene expression modeling, and Seurat/edgeR workflows. Passionate about decoding brain function through computational modeling.

Skills

- R programming
- Python
- Variant interpretation
- Single-cell RNA sequencing
- Trajectory analysis
- Agarose gel electrophoresis
- Solid-phase peptide synthesis
- Cell culture
- qPCR
- Excel
- Word
- PowerPoint
- Data entry
- Laboratory safety

Research Experience

University of Maryland, Baltimore | Baltimore, Maryland

Laboratory Intern

02/2025 - 04/2025

- Gained practical experience in AAV production, vector construction, CRISPR-based gene editing, gene therapy methodologies, and aseptic cell culture techniques.
- Engineered plasmid vectors, produced and quantified AAV vectors, and executed CRISPR-based gene knockouts, base editing, and prime editing.
- Cleaned and prepared lab equipment for solutions, specimens and samples.
- Supervised supplies in inventory and notified supervisor of low stock levels.
- Kept lab space and equipment clean to prevent contamination with hazardous substances.

Seth Ament Lab | Maryland, MD

Research Student

04/2024 - Current

- Conducted differential gene expression analysis for Huntington's disease (HD) Q111 mouse model using Pseudobulk analysis by Edge R, identifying molecular changes linked to neurodegeneration, cellular stress, and synaptic activity
- Performed cell type-specific analysis of neuronal and glial subtypes with top marker genes, ensuring accurate classification of cellular populations
- Generated UMAP dot plots, heatmaps, and volcano plots to visualize and communicate cell type-specific gene expression patterns by seurat.
- Designed and implemented data interpretation strategies to uncover cellular and molecular mechanisms underlying neurodegeneration.
- Did GSEA on DEGs to verify the genes we found in our data set.
- Identified and resolved data issues to drive accuracy and integrity.

Rajiv Gandhi Centre for Biotechnology | Thiruvananthapuram, Kerala

- Company Overview: Cancer Research
- Developed Sorafenib-loaded PLGA nanoparticles characterized

Undergraduate Student Researcher
01/2023 - 04/2023

nanoparticle size, morphology, and drug loading
- Decorated nanoparticle with TKD peptide for active targeting, confirmed coupling of TKD peptide using FTIR
- Suggested effectiveness as a targeted drug delivery system
- Solid-phase peptide synthesis (SPPS): Expertise in designing, synthesizing, and characterizing peptides
- Cancer Research

Department of Genetic Engineering, SRM University | Chennai, Tamil Nadu
Undergraduate Student Researcher
07/2022 - 11/2022

- Company Overview: Nanotechnology
- Designed and developed an efficient drug delivery system using calcium carbonate ($CaCO_3$) nanoparticles
- Utilized biocompatible nanoparticles synthesized from eggshells for enhanced biocompatibility and sustainability
- Characterized nanoparticles for drug loading efficiency and release kinetics to optimize therapeutic potential
- Nanotechnology

VJ Biotechnology
Intern
08/2021 - 08/2021

- Acquired knowledge in rDNA technology, DNA isolation, and gene sequence for genetic analysis and manipulation

Cambrionics Life Sciences
Intern
09/2019 - 09/2019

- Examined crime scenes using criminal psychology methods, including fingerprinting and blood splatter analysis
- Gained foundational knowledge of crime scene investigation techniques and their application in forensic analysis

Work Experience

Braxton Mitchell Lab - Lab Assistant
Maryland, MD
10/2024 - Current
- Managed biorepository operations, including organization, retrieval, and storage of protein samples for multiple research studies
- Maintained accurate records of sample locations and metadata using Microsoft Excel, ensuring efficient inventory tracking and accessibility
- Prepared and organized racks of protein samples for downstream aliquoting and experimental workflows in collaborating laboratories
- Ensured proper handling and storage of biological specimens to maintain sample integrity and compliance with lab protocols
- Supported laboratory operations by streamlining side tasks, contributing to the smooth execution of research projects
- Handled administrative tasks for laboratory staff and overall operations. Produced and submitted laboratory documents and reports per compliance requirements

URecFit And Wellness - University Of Maryland - Fitness Assistant
Baltimore, MD
01/2024 - 10/2024
- Provided first aid and CPR support to ensure the safety and well-being of facility patrons
- Communicated effectively with members to address inquiries, provide guidance and promote fitness programs
- Assisted in managing team operations to maintain a clean, organized, and safe workout environment
- Responded promptly to emergencies, adhering to established safety protocols and procedures

Education

University of Maryland | Baltimore, MD
Master's in Human Genetics And Genomic Medicine
05/2025

Courses: Human Genetics I and II, Biostatistics for health professional, Clinical Genetics, Genomic Applications, Genomics and bioinformatics, Programming for Bioinformatics, Research Ethics

SRM Institute of Science and Technology | Chennai, Tamil Nadu
Bachelor of Technology in Biotechnology with specialization in Genetic Engineering
04/2023
Courses: Cell biology, Molecular Biology, Human Genetics, Functional Genomics, Plant Virology.

Presentations and Conferences

1. **47th Annual UMB Graduate Research Conference** Baltimore, MD
Oral presenter April 28th 2025
Presented on “Single-nuclei RNA sequencing provides insights into the molecular signatures of the Huntington's disease mutation in the mouse striatum the therapeutic potential of an HTT-lowering antisense oligonucleotide”
2. SRM institute of Science and Technology, department of Genetic Engineering – School of Bioengineering Chennai, India
Poster presentation May 12th 2023
Presented on “Sorafenib Loaded PLGA surface modified with TKD peptide for active targeting MEM HSP70 expressing breast cancer”

Leadership and Activities

1. **UMB Indian Association** Baltimore, MD
Cultural Secretary August, 2023